R means red; B means blue. A machine has one start circle and a fixed YES or NO on each circle. Each circle has exactly one R arrow and one B arrow. Read left to right with one marker, hiding used cards. Only the current circle and the card being read may affect the next move.

Problem 5: Draw a machine that says YES exactly when its red cards can be put in groups of three with none left over. A row with no red cards must give YES. Your machine must work for rows of any length. Use as few circles as you can, and explain why fewer circles cannot work.

Problem 6: Draw a machine that says YES exactly when the last two cards are red followed by blue. Every other row must give NO. Use as few circles as you can, and explain why fewer circles cannot work.